

YASH RAWAL

+91-9399159685 | yash.mahesh.rawal@gmail.com | [linkedin.com/in/rawal-yash](https://www.linkedin.com/in/rawal-yash) | github.com/YashRL | Ratlam, MP, India

EXPERIENCE

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer

Surat, India
Oct. 2024 – Present

- Deployed enterprise agentic platform serving 1M+ active ERP users across 120+ cloud tenants.
- Architected multi-agent routing with LangGraph & SAP Joule 2.0, discovering 650+ dynamic MCP tools.
- Re-architected multi-agent execution with capability contracts, yielding 3× throughput & \$120K+ savings.
- Fine-tuned & served Qwen2.5-7B and E5-large-v2 on NVIDIA GPUs using PyTorch, Unsloth, and vLLM.
- Built 4-level hierarchical RAG in Python indexing 10K+ technical documents via PGVector (90%+ Recall@10).
- Built Prompt Lab evaluation control plane to monitor LLM outputs and curate RLHF benchmark datasets.

Bigscal Technologies Pvt. Ltd.

AI/ML Intern

Surat, India
Apr. 2024 – Sep. 2024

- Engineered clinical NLP pipelines for 116K+ medical records using 4-bit Llama 3 and FastAPI endpoints.
- Built real-time WebSocket audio streaming for Indic ASR using NVIDIA NeMo (RNN-T/CTC) and DSP.

PROJECTS

vLLM High-Performance Inference Engine | *Python, PyTorch, Pydantic, AsyncIO*

Open Source

- Merged PR #55475 fixing XML tool parser drops on `/v1/chat/completions` with torchcodecs guards.

Job Architecture Generator (Qwen2.5-7B SFT) | *Python, PyTorch, Unsloth, vLLM*

LLM Training

- Fine-tuned Qwen2.5-7B on 10K+ domain examples using Unsloth for low-latency OpenAI endpoints.

E5-large-v2 Job & Skills Embedding Fine-Tuning | *Python, PyTorch, Transformers*

Model Training

- Fine-tuned E5-large-v2 (335M) with triplet loss, boosting semantic relevance by 18% over OpenAI.

Modular Agent Harness & Conversational Copilot | *Python, FastAPI, pgvector*

Project

- Architected modular Python agent framework implementing ReAct loops with step budgets and tool routing.

Hierarchical Document Intelligence & RAG Platform | *Python, pgvector, GROBID*

Project

- Engineered 4-level hierarchical RAG ingestion (paragraphs, headings) with GROBID and PGVector retrieval.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, FastAPI, PostgreSQL, Oracle, REST APIs, WebSockets, AsyncIO, Linux

AI & ML Engineering: LLMs, Agentic AI, MCP, Tool RAG, SFT/LoRA, vLLM Serving, RLHF, Prompt Tuning

Frameworks & Libraries: LangChain, LangGraph, PyTorch, Hugging Face, vLLM, SAP Joule 2.0, NVIDIA NeMo

Cloud & DevOps: Azure, AWS, Docker, Kubernetes, CI/CD (GitHub Actions), Redis, PGVector, LLM Evaluation

EDUCATION & CERTIFICATIONS

Parul University

Master of Computer Applications (Artificial Intelligence & Machine Learning)

Gujarat, India
2022 – 2024

Certifications: Azure AI Fundamentals (AI-900) | Azure Fundamentals (AZ-900)